

## Problem Set 2 Code

● Graded

### Student

Jinku Jung

### Total Points

15 / 15 pts

### Autograder Score

15.0 / 15.0

### Passed Tests

Car constructor properly initializes value (1/1)

Car addGas() works (1/1)

Car drive() works (1/1)

The Car class compiles with the tester (2/2)

Circuit toggling both switches work (1/1)

Circuit constructor properly initializes value (1/1)

Circuit toggleSecondSwitch() works (1/1)

Circuit toggleFirstSwitch() works (1/1)

The Circuit class compiles with the tester (1/1)

The RangeInput class compiles with the tester (2/2)

increments down properly (1/1)

Range Input constructor properly initializes currentValue (1/1)

increments up properly (1/1)

### Question 2

#### Autograder Adjustments

0 / 0 pts

✓ + 0 pts Autograding Correct

### Autograder Results

Car constructor properly initializes value (1/1)

Car addGas() works (1/1)

Car drive() works (1/1)

The Car class compiles with the tester (2/2)

Circuit toggling both switches work (1/1)

Circuit constructor properly initializes value (1/1)

Circuit toggleSecondSwitch() works (1/1)

Circuit toggleFirstSwitch() works (1/1)

The Circuit class compiles with the tester (1/1)

The RangeInput class compiles with the tester (2/2)

increments down properly (1/1)

Range Input constructor properly initializes currentValue (1/1)

increments up properly (1/1)

Submitted Files

```
1  /*
2   * This is the Car class that you have to create
3   * for problem E3.13
4   *
5   * name: Jinku Jung
6   * UNI: dj2598
7   *
8   */
9
10 public class Car{
11
12     // for this problem don't worry about the maximum amount of gas
13     // allowed in the tank or tracking the overall miles driven
14
15
16     private double gas;
17     private double gasMileage;
18
19
20     public Car(double efficiency){
21         gas = 0;
22         gasMileage = efficiency;
23         // this is the constructor
24         // your code here
25
26     }
27
28     public void drive(double miles){
29         if(gasMileage * gas >= miles) {gas -= (miles / gasMileage);}
30         // this is a mutator method
31         // that reduces the gas in the tank
32         // your code here
33
34     }
35
36     public void addGas(double gasAmount){
37         gas += gasAmount;
38         // this is a mutator method
39         // that adds gas to the tank
40         // your code here
41
42     }
43
44
45     public double gasLeft(){
46         return gas;
47         // this is an accessor method for the gas in the tank
48         // your code here
49     }
```

```
50 }  
51  
52
```

▼ CarTester.java

 Download

```
1  /*  
2   * This is the Car test class  
3   * Your code for problem 3.13 must work with this  
4   * You should not change this class  
5   *  
6   *  
7   */  
8  
9  public class CarTester{  
10  
11     public static void main(String[] args){  
12         Car myCar = new Car(30.0);  
13         System.out.println("fresh car");  
14         System.out.println("current gas: " + myCar.gasLeft());  
15         System.out.println();  
16  
17         myCar.addGas(20.0);  
18         System.out.println("added 20 gallons");  
19         System.out.println("current gas: " + myCar.gasLeft());  
20         System.out.println();  
21  
22         myCar.drive(120.0);  
23         System.out.println("drove 120 miles");  
24         System.out.println("current gas: " + myCar.gasLeft());  
25  
26  
27     }  
28 }  
29
```

```
1  /*
2   * This is the Circuit class that you have to create
3   * for problem E3.4
4   *
5   * name: Jinku Jung
6   * UNI: dj2598
7   *
8   */
9
10 public class Circuit{
11
12     private int firstSwitchState;
13     private int secondSwitchState;
14     private int lampState;
15
16     public Circuit(){
17         firstSwitchState = 0;
18         secondSwitchState = 0;
19         lampState = 0;
20         // this is the constructor
21         // initialize all switches to down and light to off
22         // your code here
23
24     }
25
26     public void toggleFirstSwitch(){
27         firstSwitchState = (firstSwitchState + 1) % 2;
28         lampState = (lampState + 1) % 2;
29         // this is a mutator method
30         // that toggles the first switch
31         // your code here
32
33     }
34
35     public void toggleSecondSwitch(){
36         secondSwitchState = (secondSwitchState + 1) % 2;
37         lampState = (lampState + 1) % 2;
38         // this is a mutator method
39         // that toggles the second switch
40         // your code here
41
42     }
43
44     public int getLampState(){
45         return lampState;
46         // this is an accessor method for the lamp state
47         // your code here
48
49     }
```

```
50     }
51
52     public int getFirstSwitchState(){
53         return firstSwitchState;
54         // this is an accessor method for the first switch
55         // your code here
56     }
57
58
59     public int getSecondSwitchState(){
60         return secondSwitchState;
61         // this is an accessor method for the second switch
62         // your code here
63
64     }
65
66 }
```

---

```
1  /*
2   * This is the test class fo the Circuit class
3   * Your code for problem 3.4 must work with this
4   * You should not change this class
5   *
6   *
7   */
8
9  public class CircuitTester{
10
11     public static void main(String[] args){
12         Circuit myCircuit = new Circuit();
13
14         // test initial state
15         System.out.println("current lamp state: " + myCircuit.getLampState());
16         System.out.println("first switch state: " + myCircuit.getFirstSwitchState());
17         System.out.println("second switch state: " + myCircuit.getSecondSwitchState());
18         System.out.println();
19
20         // test first switch
21         System.out.println("Toggle first switch");
22         myCircuit.toggleFirstSwitch();
23         System.out.println("current lamp state: " + myCircuit.getLampState());
24         System.out.println("first switch state: " + myCircuit.getFirstSwitchState());
25         System.out.println("second switch state: " + myCircuit.getSecondSwitchState());
26         System.out.println();
27
28         // test second switch
29         System.out.println("Toggle second switch");
30         myCircuit.toggleSecondSwitch();
31         System.out.println("current lamp state: " + myCircuit.getLampState());
32         System.out.println("first switch state: " + myCircuit.getFirstSwitchState());
33         System.out.println("second switch state: " + myCircuit.getSecondSwitchState());
34         System.out.println();
35
36         // test first switch again
37         System.out.println("Toggle first switch");
38         myCircuit.toggleFirstSwitch();
39         System.out.println("current lamp state: " + myCircuit.getLampState());
40         System.out.println("first switch state: " + myCircuit.getFirstSwitchState());
41         System.out.println("second switch state: " + myCircuit.getSecondSwitchState());
42         System.out.println();
43
44     }
45
46
47
48 }
```

```
1  /*
2   * This is the RangeInput class that you have to create
3   * for problem E3.3
4   *
5   * name: Jinku Jung
6   * UNI: dj2598
7   *
8   */
9
10 public class RangeInput{
11
12     private int currentValue;
13     private int highValue;
14     private int lowValue;
15
16     public RangeInput(int high, int low){
17         currentValue = 0;
18         highValue = 0;
19         lowValue = 0;
20
21         highValue += high;
22         lowValue += low;
23         currentValue = lowValue + (int) (highValue - lowValue)/2;
24         // this is the constructor
25         // your code here
26
27     }
28
29     public void up(){
30         if (currentValue < highValue) {currentValue++;}
31         // this is a mutator method
32         // that increases currentValue if allowed
33         // your code here
34
35     }
36
37     public void down(){
38         if(currentValue > lowValue) {currentValue--;}
39         // this is a mutator method
40         // that decreases currentValue if allowed
41         // your code here
42
43     }
44
45     public int getCurrentValue(){
46         return currentValue;
47         // this is an accessor method
48         // your code here
49     }
```

```
50
51     }
52
53
54
55 }
```

▼ RangeInputTester.java

 Download

```
1  /*
2   * This is the RangeInput test class
3   * Your code for problem 3.3 must work with this
4   * You should not change this class
5   *
6   *
7   */
8
9  public class RangeInputTester{
10
11      public static void main(String[] args){
12          RangeInput myInput = new RangeInput(30,20);
13          System.out.println("current value: " + myInput.getCurrentValue());
14          for (int i=0;i<4;i++){
15              myInput.up();
16          }
17          System.out.println("new value: " + myInput.getCurrentValue());
18
19          for (int i=0;i<50;i++){
20              myInput.down();
21          }
22          System.out.println("last value: " + myInput.getCurrentValue());
23      }
24
25
26  }
```